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EXPERIMENT 7

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Subject Name: Computer Network

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Experiment

VLAN Configuration, Router-on-a-Stick & Multi-VLAN DHCP Server

Aim

To configure a router to act as a DHCP server for multiple VLANs, using Cisco Packet Tracer.

Learning Objectives

After completing this experiment, I should be able to:

1. Understand the concept of VLANs.
2. Create and configure multiple VLANs on a switch.
3. Assign switch ports to the appropriate VLANs.
4. Distinguish between an Access Port and a Trunk Port.
5. Configure a trunk between a switch and a router.
6. Understand Inter-VLAN Routing and the Router-on-a-Stick technique.
7. Configure router subinterfaces for multiple VLANs.
8. Configure a router as a DHCP Server with a separate pool per VLAN.
9. Verify the VLAN, trunk, DHCP, and routing configuration.

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Theory

VLANs and Broadcast Domains

A VLAN (Virtual Local Area Network) is a logical network carved out of a physical switch. Rather than treating every connected device as part of one broadcast domain, a switch's ports can be grouped into separate VLANs — for instance, one VLAN for one group of hosts and another for a second group — even though all the devices sit on the same physical switch. This gives better network segmentation, tighter security, smaller broadcast domains per group, and easier management overall.

Inter-VLAN Routing and Router-on-a-Stick

Because each VLAN is its own Layer 2 network, a switch by itself cannot move traffic from one VLAN to another — a Layer 3 device (a router, or a Layer 3 switch) is needed, a process known as Inter-VLAN Routing. The Router-on-a-Stick technique achieves this using a single physical router interface split into several logical subinterfaces, one per VLAN, so one cable and one port can service every VLAN instead of needing a dedicated link per VLAN. Each subinterface gets its own IP address, which becomes the default gateway for that VLAN.

VLAN Tagging (IEEE 802.1Q)

When multiple VLANs share a single trunk link, the device on the receiving end needs a way to tell which VLAN each frame came from. This is handled by 802.1Q VLAN tagging, which stamps a VLAN identifier onto the Ethernet frame as it crosses the trunk. On the router side, the command `encapsulation dot1Q <vlan-id>` tells a given subinterface which VLAN's tagged frames it should process.

DHCP and the DORA Process

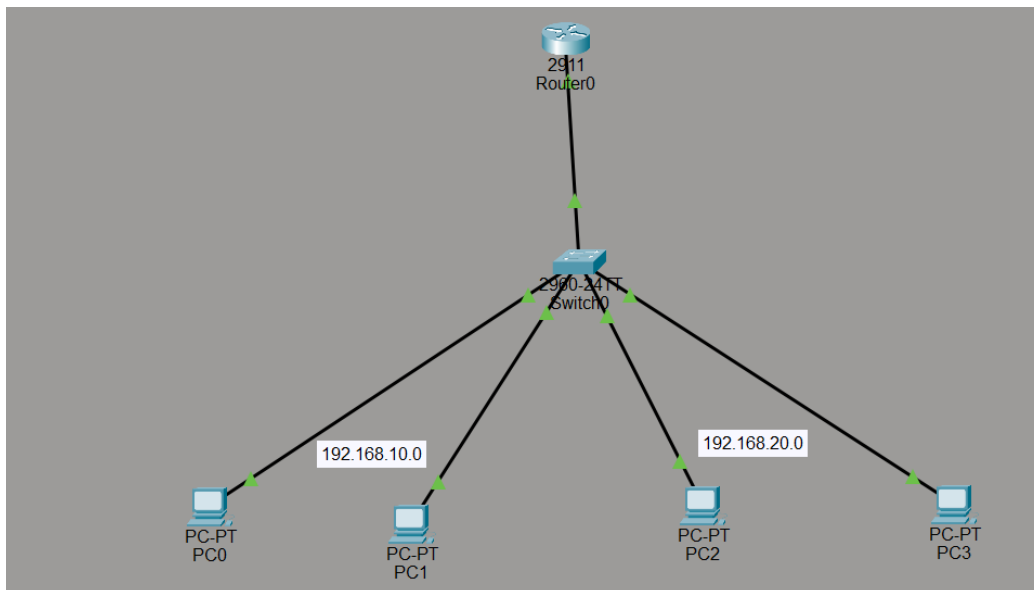
DHCP (Dynamic Host Configuration Protocol) automatically hands out network configuration — IP address, subnet mask, default gateway, and DNS server — to client devices, so nobody has to configure each host by hand. A Cisco router can act as its own DHCP server. The exchange happens in four steps, remembered as DORA:

- Discover — the client broadcasts a request asking if any DHCP server is available.
- Offer — a server responds with an available IP address.
- Request — the client asks to use the offered address.
- Acknowledgment — the server confirms the allocation.

When a single router serves more than one VLAN, it keeps a separate DHCP pool per VLAN, each tied to that VLAN's own subnet, so VLAN 10 clients draw addresses from the VLAN10 pool and VLAN 20 clients draw addresses from the VLAN20 pool. The address already in use by the router's own subinterface (the gateway) is marked with `ip dhcp excluded-address` so DHCP never hands it out to a client by mistake.

Network Topology

Router0 connects down to Switch0 over a single trunk link carrying both VLANs; the switch's four remaining ports are split two-and-two between the two VLANs — PC0 and PC1 sit on the 192.168.10.0/24 side, PC2 and PC3 sit on the 192.168.20.0/24 side. The screenshot below shows this topology as built in Cisco Packet Tracer, on top of which the router-on-a-stick and DHCP configuration described below was layered.



Devices Used

1 Router (2911) — Router0; 1 Switch (2960-24TT) — Switch0; 4 PCs — PC0, PC1, PC2, PC3. All PC-to-switch and switch-to-router links use Copper Straight-Through cables.

VLAN and Addressing Plan

VLAN	Name	Network	Gateway
10	VLAN10	192.168.10.0/24	192.168.10.1
20	VLAN20	192.168.20.0/24	192.168.20.1

Device / Port / VLAN Mapping

Device	Switch Port	VLAN
PC0	Fa0/1	VLAN 10
PC1	Fa0/2	VLAN 10
PC2	Fa0/3	VLAN 20
PC3	Fa0/4	VLAN 20
Router0	Fa0/24	Trunk (802.1Q)

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Procedure

Part A — Configure the Switch (Switch0)

Step 1 — Enter privileged and global configuration mode

```
Switch> enable
Switch# configure terminal
Switch(config)#
```

Step 2 — Create VLAN 10 and VLAN 20

```
Switch(config)# vlan 10
Switch(config-vlan)# name VLAN10
Switch(config-vlan)# exit

Switch(config)# vlan 20
Switch(config-vlan)# name VLAN20
Switch(config-vlan)# exit
```

Step 3 — Confirm the VLANs were created

```
Switch(config)# do show vlan brief
```

Step 4 — Assign PC0 and PC1 (Fa0/1–Fa0/2) to VLAN 10

```
Switch(config)# interface range fastEthernet 0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

Step 5 — Assign PC2 and PC3 (Fa0/3–Fa0/4) to VLAN 20

```
Switch(config)# interface range fastEthernet 0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

Step 6 — Configure Fa0/24 as a trunk port (link to the router)

```
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

Step 7 — Confirm the trunk

```
Switch# show interfaces trunk
```

Fa0/24 should now show up as a trunk port, confirming the switch is ready to carry both VLANs toward the router.

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Part B — Configure the Router (Router0)

Step 1 — Enter privileged and global configuration mode

```
Router> enable
Router# configure terminal
Router(config)#
```

Step 2 — Bring up the physical interface G0/0

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Step 3 — Create the subinterface for VLAN 10

```
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
```

Step 4 — Create the subinterface for VLAN 20

```
Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

Step 5 — Verify the router interfaces

```
Router# show ip interface brief
```

Both G0/0.10 and G0/0.20 should show as up/up, holding 192.168.10.1 and 192.168.20.1 respectively. This is Router-on-a-Stick: one physical interface, two logical subinterfaces, each acting as the gateway for its own VLAN, reached over a single trunk link to the switch.

Step 6 — Exclude the gateway addresses from DHCP

```
Router(config)# ip dhcp excluded-address 192.168.10.1
Router(config)# ip dhcp excluded-address 192.168.20.1
```

Step 7 — Create the DHCP pool for VLAN 10

```
Router(config)# ip dhcp pool VLAN10_POOL
Router(dhcp-config)# network 192.168.10.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

Step 8 — Create the DHCP pool for VLAN 20

```
Router(config)# ip dhcp pool VLAN20_POOL
Router(dhcp-config)# network 192.168.20.0 255.255.255.0
```

```
Router(dhcp-config)# default-router 192.168.20.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# exit
```

Part C — Configure the PCs

On each PC, open Desktop → IP Configuration and select DHCP instead of a static address. PC0 and PC1 (VLAN 10) should automatically pick up addresses in 192.168.10.0/24, while PC2 and PC3 (VLAN 20) should pick up addresses in 192.168.20.0/24.

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Verification / Output

show vlan brief (on Switch0)

VLAN	Name	Status	Ports
1	default	active	
10	VLAN10	active	Fa0/1, Fa0/2
20	VLAN20	active	Fa0/3, Fa0/4

show interfaces trunk (on Switch0)

Port	Mode	Encapsulation	Status	Native vlan
Fa0/24	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Fa0/24	1,10,20			

show ip interface brief (on Router0)

Interface	IP-Address	Status	Protocol
GigabitEthernet0/0	unassigned	up	up
GigabitEthernet0/0.10	192.168.10.1	up	up
GigabitEthernet0/0.20	192.168.20.1	up	up

IP Configuration received by each PC (via DHCP)

PC	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

Note: the exact addresses handed out can shift slightly depending on lease order in Packet Tracer — what matters is that each PC lands in the correct VLAN's subnet with the correct gateway.

show ip dhcp binding (on Router0)

IP address	Client-ID/Hardware address	Lease expiration
192.168.10.2	0100.5079.6668.0a	...
192.168.10.3	0100.5079.6668.0b	...
192.168.20.2	0100.5079.6668.0c	...
192.168.20.3	0100.5079.6668.0d	...

Result

Switch0 was configured with two VLANs — VLAN 10 and VLAN 20 — connected to Router0 over a single trunk link. Using Router-on-a-Stick, the router's one physical interface was split into two subinterfaces, each serving as the default gateway for its own VLAN. The router was then set up as a DHCP server with a separate address pool per VLAN, so PC0/PC1 automatically picked up addresses in 192.168.10.0/24 and PC2/PC3

automatically picked up addresses in 192.168.20.0/24 — confirming that one router can provide both inter-VLAN routing and DHCP service to multiple VLANs at the same time.

Learning Outcome

After completing this experiment, the following learning outcomes were achieved:

- Understood how VLANs logically split a single physical switch into separate broadcast domains.
- Learned to create VLANs and assign access ports to them using switchport access vlan.
- Understood the difference between an access port (single VLAN, untagged) and a trunk port (multiple VLANs, 802.1Q-tagged).
- Implemented Router-on-a-Stick using subinterfaces and encapsulation dot1Q to perform inter-VLAN routing over one physical link.
- Configured a Cisco router as a DHCP server with per-VLAN address pools, excluded gateway addresses, and default-router/DNS options.
- Verified the configuration end-to-end using show vlan brief, show interfaces trunk, show ip interface brief, and show ip dhcp binding.
- Gained hands-on experience building and testing a multi-VLAN, DHCP-enabled network in Cisco Packet Tracer.